

Kubernetes Error 06 : Pod Stuck in Terminating State

What is a Pod Stuck in Terminating State?

- A pod enters the Terminating state when Kubernetes is trying to delete it, but the deletion does not complete successfully.
- Common causes include applications that don't exit gracefully, volume unmount failures, blocking finalizers, container runtime or kubelet issues, and long termination grace periods.
- The first step is to inspect the Pod with `kubectl describe pod`, and force deletion should only be used as a last resort.  

Kubernetes Pod Stuck in Terminating State — Common Reasons & Solutions

 Cause	 Description	 Solution	 Useful Commands
 Hanging Application Process	Application does not exit after receiving the SIGTERM signal	Fix graceful shutdown logic and ensure the application exits cleanly	<code>kubectl logs <pod-name></code>
 Volume Unmount Issues	Persistent Volumes (PV/PVC) cannot be detached or unmounted	Verify PVC, PV, StorageClass, CSI Driver, and volume attachments	<code>kubectl get pvc</code> <code>kubectl describe pvc <pvc-name></code>
 Finalizers	Kubernetes finalizers block Pod deletion until cleanup completes	Check and remove stale finalizers only after confirming cleanup is safe	<code>kubectl get pod <pod-name> -o yaml</code>

 Cause	 Description	 Solution	 Useful Commands
 Container Runtime Issues	containerd or Docker is unresponsive	Restart the container runtime and inspect its logs	<code>systemctl status containerd</code>
 API Server or Kubelet Issues	Communication problems delay Pod deletion	Verify kubelet status and API server connectivity	<code>systemctl status kubelet</code>
 Long terminationGracePeriodSeconds	Kubernetes waits before forcefully terminating the Pod	Reduce the grace period if appropriate or use force deletion only when necessary	Check Pod specification

Check the Pod Status

```
kubectl get pods
```

NAME	READY	STATUS	AGE
nginx-abc	0/1	Terminating	10m

Pod Terminating Troubleshooting Flow

```
Pod Status = Terminating
↓
kubectl describe pod <pod-name>
↓
Check Events
↓
Application Hanging?
↓
Volume Unmount Issue?
↓
Finalizers Present?
↓
Kubelet / Container Runtime Healthy?
↓
Force Delete (if required)
```

↓
Adjust terminationGracePeriodSeconds
↓
PodDeleted Successfully ✅

Useful Troubleshooting Commands

 Command	 Purpose
<code>kubectl get pods</code>	Check pod status
<code>kubectl describe pod <pod-name></code>	View events and termination details
<code>kubectl get pod <pod-name> -o yaml</code>	Check finalizers and metadata
<code>kubectl delete pod <pod-name> --grace-period=0 --force</code>	Force delete a stuck pod
<code>kubectl describe pvc</code>	Check Persistent Volume Claim status
<code>kubectl get pv</code>	View Persistent Volumes
<code>kubectl describe node <node-name></code>	Check node health
<code>kubectl get events --sort-by=.metadata.creationTimestamp</code>	View recent cluster events
<code>kubectl logs <pod-name></code>	Check application shutdown logs
<code>systemctl status kubelet</code>	Verify kubelet is running
<code>systemctl status containerd</code>	Verify container runtime status

Interview Answer

Q: How do you troubleshoot a Pod stuck in the Terminating state?

- Run `kubectl describe pod <pod-name>` to inspect events.
- Check whether the application exits gracefully after receiving `SIGTERM`.
- Verify that PVs/PVCs are detached and unmounted correctly.
- Check the health of the kubelet and container runtime.

Use `kubectl delete pod --grace-period=0 --force` only when graceful termination is no longer possible.